

HEMANTH KUMAR C S

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SUMMARY

Final-year AI & ML Engineering student with industry experience at HAL (Hindustan Aeronautics Limited) and hands-on expertise in deep learning, computer vision, and federated learning. Built production-grade ML systems for traffic management, cybersecurity, and healthcare with real-time inference on edge devices. Strong track record of research publications (IEEE) and hackathon wins. Seeking ML Engineer / Deep Learning Engineer roles.

EDUCATION

B.E., Computer Science & Engineering (AI & ML) - ATME College of Engineering, VTU
Expected 2026 | CGPA: 8.4/10

TECHNICAL SKILLS

Programming Languages: Python, Java, C, R

ML / DL: TensorFlow, Keras, PyTorch, Flower (Federated Learning), scikit-learn, ONNX Runtime

CV: OpenCV, YOLOv8/v11, MediaPipe, ByteTrack

Models & Concepts: CNNs (ResNet, YOLO), attention/transformer concepts (familiar), classic ML

Web & DB: Flask, Streamlit, MongoDB, MySQL, Firebase, Supabase

Edge: TensorFlow Lite (INT8 quantization), Raspberry Pi, Modbus/RS485

Tools: Git, Docker, Jupyter, Power BI

EXPERIENCE

Student Intern - IT Department, Overhaul Division, HAL, Bangalore | Jan 2026 - Present

- Built a robust solution for the Customer Service department to resolve real-time inter-department communication and documentation challenges.
- Developed a robust intranet-based solution for the Overhaul division to streamline inter-department folder organization and file accessibility.

Cybersecurity Intern - Academor (Remote) | Aug 2023 - Nov 2023

- Performed network security training and hands-on vulnerability assessments

Media Head - IET onCampus ATMECE | Dec 2024 - Present

- Led content & outreach for a 60+ member society; increased event engagement by ~60%.
- Organized 7+ technical events and a 24-hour state hackathon

PROJECTS

Dynamic Traffic Emergency Control System (D-TECS)

Real-time AI Traffic Management with Emergency Vehicle Priority

- Built end-to-end computer vision pipeline using YOLOv11 + ONNX Runtime for real-time vehicle detection and density analysis at traffic intersections

- Built Emergency Priority System (EPS) that detects ambulances via visual identifiers and dynamically creates "Green Corridors" by overriding signal cycles; reduces emergency response time
- Implemented adaptive signal timing algorithms based on live lane-wise vehicle counts using realistic Indian traffic logic (clockwise N -> E -> S -> W phase control)
- Developed full-stack architecture: Python AI engine (OpenCV/ByteTrack) → WebSocket backend → React dashboard with real-time video analytics and manual override controls

Privacy-Preserving Federated Learning for DDoS Detection

IEEE ICAFT 2025 - Conference paper accepted

- Engineered 1D-CNN-based intrusion detection using Flower framework for federated learning; trains on distributed network data without centralization
- Implemented Multi-Krum + FedAvg aggregation to defend against Byzantine attacks from malicious clients
- Achieved 80% detection accuracy on CIC-DDoS2019 dataset with 30-feature network traffic schema
- Built async federated simulation with 4 clients + monitoring dashboard; containerized with Docker Compose

StreeRaksha - Real-Time Women's Safety Platform

- Built real-time threat detection pipeline on edge devices (Raspberry Pi/Jetson) using YOLOv11s + RTMPose-m + PromptPAR; 6-factor risk scoring (pose, gender, proximity, time, grouping, motion)
- Architected Go backend with WebSocket broker (Gin) handling AI stream ingestion, Firebase persistence, and live dashboard broadcast; deployed via Docker multi-stage builds (10MB binary)
- Implemented state machine alerting with 4 risk levels, evidence capture on ALERT (80+ score), and configurable thresholds for production deployment

Melanocytic Nevi Classification - IgMin Research: [Journal paper publication](#)

- Built ResNet50 classifier on 10,000+ dermoscopic images (ISIC dataset); achieved 89% accuracy (vs. 82% baseline) through class weighting and heavy augmentation
- Used class weighting + augmentation to improve generalization

AgroBalance - Smart Soil Monitoring

- Integrated NPK sensor + Raspberry Pi via Modbus RTU; built ML recommendation engine for crop/fertilizer optimization

Move Mentor - Student Bus Tracking

- Real-time tracking, role-based access, route optimization reducing average wait time of students

ACHIEVEMENTS & CERTIFICATIONS

- IEEE ICAFT 2025: Research paper selected for presentation on Federated Learning for DDoS Detection
- *Winner*: "Invaders" Hackathon, MIT Mysuru (2023)
- *Runner-up*: "Code Battle 2K25", KLSVDIT, Haliyal (2025)
- *Finalist*: RVCExIITB CTF (2024)
- *Certifications*: TensorFlow ML, Nvidia - Building LLM Apps with Prompt Engineering, Tableau, Oracle SQL